# STAFF REVIEW · H446 089 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

## Do Now — source answers

1. Holds address of next instruction.

2. Arithmetic and logical operations.

3. Address, data or control bus.

4. Q becomes1, as D is sampled at the active edge.

## I Do — source worked steps

1. Model response — The address identifies where; data is what is transferred. CU coordinates the process.

## We Do — guided CHECK

1. Model response
Saying the CU performs arithmetic confuses separate roles.

## You Do — source mark schemes

1. ALU arithmetic/logic [2], CU manages decoding/control [2], registers fast working storage [2].

2. Address bus carries requested memory location [2], control bus read request [2], data bus carries returned instruction/data bits [2].

3. Supports separate roles/paths [2], names correctly map [2], warns shared memory/bus contention and simplification of complex physical hardware [2].

## Exit Ticket — source answer

1. False. Address bus carries the location; data bus carries contents.